Calculus and Ordinary Differential Equations

AS1109 – Section H

Accelerated Student Notes: Lessons 9–15 (Mid-Sem Completion Block)

How to use this handout. Definitions, formulas, representative examples and difficult diagrams are already printed. Use the blank spaces for short explanations, calculations, questions and extra examples from class. You are not expected to copy the printed material from the board.

Lesson	Main content	Thomas–Finney anchor
9	Differentiability; tangent-plane linearisation	§12.4, pp. 933–936
10	Differentials; sensitivity; error estimation	§12.4, pp. 937–943
11	Multivariable Chain Rule; composite variables	§12.5
12	Local/absolute extrema; saddle points; second derivative test	§12.8
13	Engineering optimisation; arc length of plane curves	§12.8 + §5.5
14	Cylinders and quadric surfaces: recognition and sketching	§10.6
15	Solids of revolution: surface area and volume	§5.6 + §5.3

This block follows the actual Section-H progress rather than the administrative lecture number. The official AS1109 syllabus requires differentiability/total derivative, linearisation, Chain Rule, extrema/saddle points, arc length, quadric surfaces, and solids of revolution before the Mid-Sem boundary.

Lesson 9: Differentiability and linearisation

Learning goals. Distinguish existence of partial derivatives from differentiability; use the increment form; form the tangent-plane/linearisation L.

1. What does "differentiable" mean?

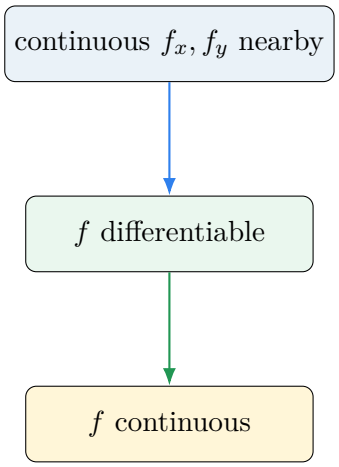
For $z = f(x, y)$ near (x_0, y_0), write

$$\Delta x = x - x_0, \qquad \Delta y = y - y_0, \qquad \Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0).$$

Thomas–Finney increment form. If f is differentiable at (x_0, y_0), then

$$\Delta z = f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y + \varepsilon_1 \Delta x + \varepsilon_2 \Delta y,$$

where $\varepsilon_1, \varepsilon_2 \to 0$ as $(\Delta x, \Delta y) \to (0, 0)$. The first two terms are the useful *linear* part; the remaining terms are smaller-order errors.



Useful sufficient condition: continuous first partials in an open region ⇒ differentiability there.

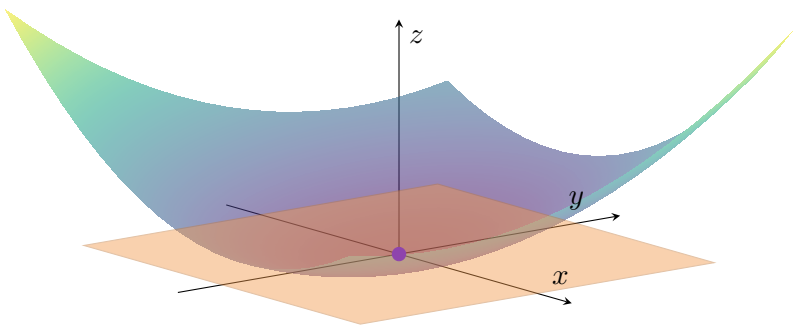
Important: merely having f_x, f_y at one point does *not* force differentiability.

2. Linearisation = tangent-plane model

At (x_0, y_0),

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

For (x, y) close to (x_0, y_0), $f(x, y) \approx L(x, y)$.



At the contact point the tangent plane and the surface agree in value and first-order slopes. Curvature produces the remaining error farther away.

Example. Let $f(x,y) = \sqrt{4 - x^2 - y^2}$. At $(0,0)$,

$$f(0,0) = 2, \quad f_x(0,0) = 0, \quad f_y(0,0) = 0,$$

so $L(x,y) = 2$. Near the top of the upper hemisphere, the tangent plane is horizontal.

Exam-style 3-variable extension. For $f(x,y,z) = \ln(x + yz)$ at $P = (2,1,-1)$,

$$L = f(P) + f_x(P)(x-2) + f_y(P)(y-1) + f_z(P)(z+1).$$

Compute the three coefficients in class.

Concept check. Existence of $f_x(a,b)$ and $f_y(a,b)$ alone is not enough to claim differentiability. A common safe test in routine problems is: first partial derivatives exist and are continuous in a neighbourhood $\Rightarrow$ differentiable.

Lesson 10: Differentials, sensitivity and error estimation

Learning goals. Use the total differential to estimate small output changes; interpret sensitivity coefficients; estimate absolute/relative/percentage error.

For $z = f(x, y)$,

$$dz = df = f_x\,dx + f_y\,dy, \qquad \Delta f \approx df$$

for sufficiently small input changes. For more variables, add one partial-derivative term per independent variable.



Engineering example: cylindrical tank. $V = \pi r^2 h$ gives

$$dV = 2\pi rh\,dr + \pi r^2\,dh.$$

The coefficient $2\pi rh$ measures sensitivity to radius; πr^2 measures sensitivity to height. Dividing by V gives

$$\frac{dV}{V} = 2\frac{dr}{r} + \frac{dh}{h}.$$

This makes percentage-error propagation almost immediate.

Absolute, relative and percentage error

$$\text{absolute error} \approx |df|, \qquad \text{relative error} \approx \left|\frac{df}{f}\right|, \qquad \text{percentage error} \approx 100\left|\frac{df}{f}\right|\%.$$

Yo-yo/tension-type model. If

$$T(r, R) = \frac{mgr}{2r^2 + R^2},$$

then

$$dT = T_r\,dr + T_R\,dR.$$

Before substituting numbers, use the signs of $T_r dr$ and $T_R dR$ to predict whether the tension tends to rise or fall.

Error-bound pattern

If measurements satisfy $|dx| \le a$ and $|dy| \le b$, then a practical first-order bound is

$$|df| \le |f_x|a + |f_y|b.$$

Differentials give a first-order estimate, not an exact change. The approximation improves as the increments shrink and the function remains smooth near the operating point.

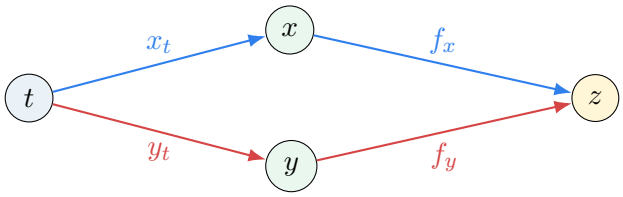
Lesson 11: The multivariable Chain Rule

Learning goals. Differentiate composite multivariable functions; read dependency diagrams; handle one-parameter and two-parameter compositions.

1. One parameter

If $z = f(x, y)$ with $x = x(t)$ and $y = y(t)$,

$$\frac{dz}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt}.$$



Each complete path from t to z contributes a product; add the path products.

If $z = e^{xy}$, $x = r\cos\theta$ and $y = r^2\sin\theta$, then

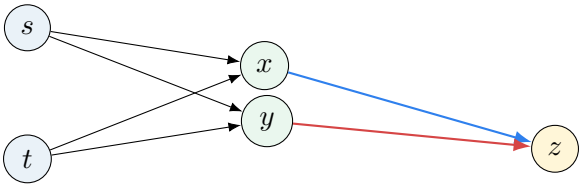
$$\frac{\partial z}{\partial r} = z_x \frac{\partial x}{\partial r} + z_y \frac{\partial y}{\partial r}.$$

This is exactly the dependency-tree method.

2. Two parameters

If $z = f(x, y)$ and $x = x(s, t)$, $y = y(s, t)$,

$$z_s = f_x x_s + f_y y_s, \qquad z_t = f_x x_t + f_y y_t.$$



3. Physical rate example

If $T(x, y)$ is a temperature field and a robot follows $(x(t), y(t))$, then

$$\frac{dT}{dt} = T_x x'(t) + T_y y'(t).$$

So the robot's sensed heating/cooling rate depends both on the field and on how the robot moves through it.

When a question gives several layers of variables, draw the dependency graph first. It prevents missing terms.

Lesson 12: Extreme values, saddle points and the second derivative test

Learning goals. Find critical points; classify local extrema/saddles; distinguish local from absolute extrema; examine boundaries of closed regions.

1. Critical points

For an interior point of a differentiable function, a local maximum/minimum can occur only where

$$f_x = 0, \qquad f_y = 0.$$

Such points are **critical points**. Critical points may be maxima, minima, saddles, or inconclusive cases.

2. Second derivative test

At a critical point (a, b) define

$$D = f_{xx}(a,b) f_{yy}(a,b) - \left(f_{xy}(a,b)\right)^2.$$

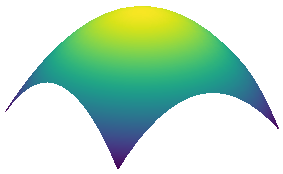
Then:

$D > 0,\ f_{xx} > 0$	local minimum,
$D > 0,\ f_{xx} < 0$	local maximum,
$D < 0$	saddle point,
$D = 0$	test inconclusive.

minimum



maximum

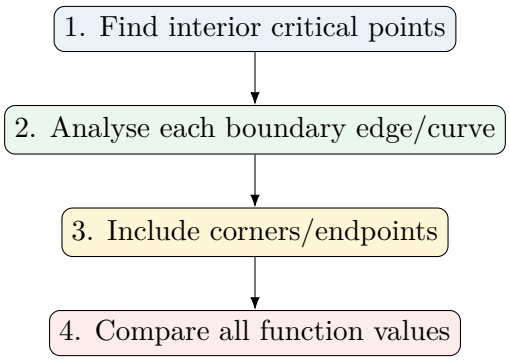


saddle



For $f(x, y) = x^2 + y^2 - 4x + 6y + 5$, solve $f_x = f_y = 0$, then classify. Completing squares gives a useful geometric check.

3. Absolute extrema on a closed bounded region



A rectangle problem is not finished after solving $f_x = f_y = 0$. Absolute extrema may lie on an edge or at a corner.

Lesson 13: Engineering optimisation and arc length

Learning goals. Translate an engineering design objective into a multivariable optimisation problem; compute arc length of a plane curve.

A. Optimisation workflow

Model variables + constraints → Objective f → Critical/boundary candidates → Compare + interpret

Engineering example. A rectangular plate of fixed area A has side lengths x, y with $xy = A$. If an edge-cost model is $C = 2c_x x + 2c_y y$, eliminate one variable using $y = A/x$, optimise the resulting one-variable function, and interpret the optimum dimensions.

B. Arc length

For a smooth graph $y = f(x)$ from $x = a$ to $x = b$,

For $y = f(x)$,

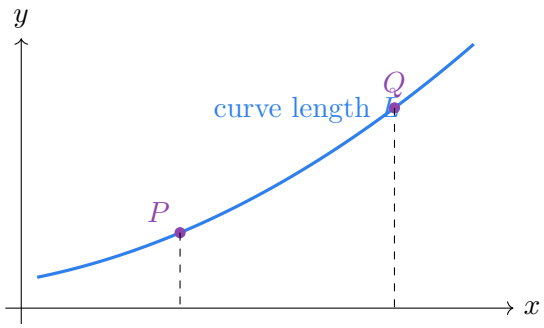
$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx.$$

If $x = g(y)$ is simpler, use

$$L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy.$$

Differential form:

$$ds = \sqrt{dx^2 + dy^2}, \qquad L = \int ds.$$



Astroid-type arc. For $x^{2/3} + y^{2/3} = 4$, choose the branch/variable that makes differentiation manageable. Symmetry can reduce the amount of integration, but the requested arc endpoints must still be respected.

Before integrating, sketch the requested arc and check whether the derivative becomes singular at an endpoint. Sometimes rewriting x as a function of y (or vice versa) avoids an awkward derivative.

Lesson 14: Cylinders and quadric surfaces

Learning goals. Recognise cylinders and standard quadrics; use traces to sketch; infer axis/orientation from the equation.

1. Cylinder test

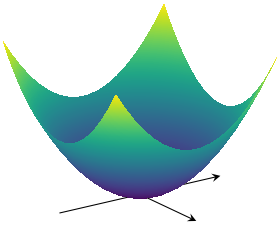
If one coordinate is missing from an equation, the surface extends parallel to that coordinate axis. Example: $y = x^2$ contains no z, so it is a parabolic cylinder parallel to the z-axis.

2. Trace method

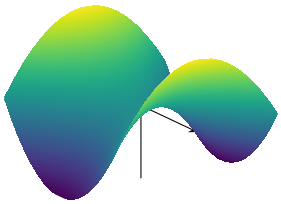
Equation → Set $x = 0$, $y = 0$, $z = 0$ → Identify 2D traces → Assemble 3D shape

Surface	Standard sign pattern	Recognition clue
Ellipsoid	$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$	all squared terms positive; bounded
Hyperboloid (1 sheet)	two +, one $-$ $= 1$	connected "waist"; axis corresponds to negative term
Hyperboloid (2 sheets)	one +, two $-$ $= 1$	two disconnected pieces; axis corresponds to positive term
Elliptic paraboloid	$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$	bowl; one unsquared variable
Hyperbolic paraboloid	$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}$	saddle
Elliptic cone	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$	double cone through origin

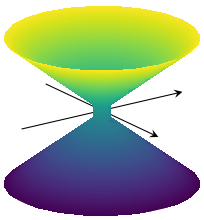
elliptic paraboloid



hyperbolic paraboloid



cone



Do not memorise pictures alone. Use traces. For example, setting $z = k$ in an elliptic paraboloid gives ellipses; setting $x = 0$ or $y = 0$ gives parabolas.

Lesson 15: Solids of revolution: surface area and volume

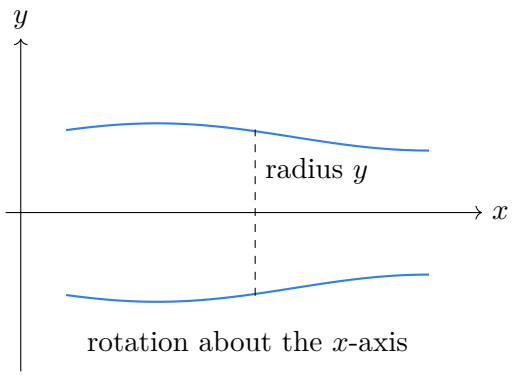
Learning goals. Choose the correct radius and differential; distinguish surface-area and volume formulas; set up disks/washers correctly.

1. Surface area of revolution

For $y = f(x) \geq 0$ revolved about the x-axis,

$$S = 2\pi \int_a^b y \, ds = 2\pi \int_a^b f(x)\sqrt{1 + [f'(x)]^2}\, dx.$$

More generally: **surface area** $= \int 2\pi(\text{radius})\, ds$.

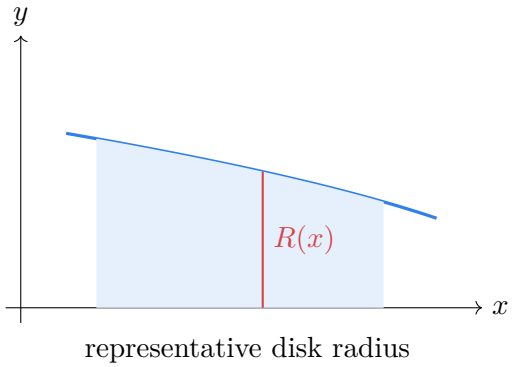


2. Volumes by disks and washers

Cross-sectional area A gives

$$V = \int A \, d(\text{axis variable}).$$

Disk: $A = \pi R^2$. Washer: $A = \pi(R^2 - r^2)$.



3. Choosing dx or dy

Use dx when	Use dy when
vertical slices are simplest; radii are functions of x solving for y is natural	horizontal slices are simplest; radii are functions of y solving for x avoids multiple branches

Method check. If the region between $y = x^2$ and $y = 4$ is revolved about the x-axis, a vertical slice gives a washer. Identify the outer and inner radii from the axis before writing the integral.

Surface area and volume can look deceptively similar. Surface area has a circumference factor $2\pi r$ multiplied by arc-length ds; disk/washer volume has a cross-sectional area $\pi(R^2 - r^2)$ multiplied by thickness.

Mid-Sem readiness checklist

By the end of Lesson 15 you should be able to:

- test/justify differentiability in routine smooth cases;
- form a linear approximation;
- estimate change using differentials;
- apply the multivariable Chain Rule;
- find/classify critical points;
- find absolute extrema on rectangles;
- calculate arc length;
- recognise/sketch standard quadrics;
- set up surface-area integrals;
- set up disk/washer volume integrals.